How does the implementation of solar systems impact the development of Central Africa?

Assumptions

When I began this project, I expected to determine that the implementation of solar systems had a positive effect on local socioeconomic states in Central Africa. However, I soon found that its effect on socioeconomic states was more complex and confounded than a simple scalar boost.

Definitions

Photovoltaics: Materials that convert solar energy into electrical energy
Central Africa: A subregion of the African continent centered primarily around the Congo Basin
Development: Measured as the score on the inequality-adjusted human development index
Solar Resource: the total amount of electromagnetic radiation emitted by the sun that reaches a specific geographic area on Earth

Perspectives

Falchetta focuses on the feasibility and economic potential of solar-based irrigation systems in Sub-Saharan Africa. It found that "irrigation could support smallholder farmers with stabilizing crop yields, increasing incomes, and achieving food security" while "a key barrier to irrigation uptake is inadequate solar electricity . . . [we devise a] modeling framework to show that over one third of unmet crop water requirements . . . could be supplied with standalone solar photovoltaic (PV) irrigation systems that can be paid back by farmers within 20 years" (Falchetta et al.).
Maqbool validates this with their results that "predicted that [socioeconomic factors] are the major challenges in hindering the overall sustainability in Africa rather than technical complexities . . . [a similar outcome] also validated by the Case study of ITC centre powered project of solar energy" (Maqbool et al.) Wassie validates these results with evidence from southern Ethiopia which suggested that "a solar-electrified rural household could save the consumption of 43.68 L of kerosene and emission of 107 kg CO_2 per year compared to a non-electrified one . . . [which could]

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Works Cited

Falchetta, Giacomo, et al. "Solar Irrigation in Sub-Saharan Africa: Economic Feasibility and Development Potential." *Environmental Research Letters*, vol. 18, no. 9, Sept. 2023, p. 094044, https://doi.org/10.1088/1748-9326/acefe5.

Fraser, Timothy. "Does Social Capital Boost or Block Renewable Energy Siting? South African Solar Politics in Comparison." *Energy Research & Social Science*, vol. 71, Jan. 2021, p. 101845, https://doi.org/10.1016/j.erss.2020.101845.

Maqbool, Rashid, and Stephen Arome Akubo. "Solar Energy for Sustainability in Africa: The Challenges of Socio-Economic Factors and Technical Complexities." *International Journal of Energy Research*, vol. 46, no. 12, July 2022, https://doi.org/10.1002/er.8425.

Szábó, Sándor, et al. "Mapping of Affordability Levels for Photovoltaic-Based Electricity Generation in the Solar Belt of Sub-Saharan Africa, East Asia and South Asia." *Scientific Reports*, vol. 11, no. 1, Feb. 2021, https://doi.org/10.1038/s41598-021-82638-x.

Wassie, Yibeltal T., and Muyiwa S. Adaramola. "Socio-Economic and Environmental Impacts of Rural Electrification with Solar Photovoltaic Systems: Evidence from Southern Ethiopia." *Energy for Sustainable Development*, vol. 60, Feb. 2021, pp. 52–66, https://doi.org/10.1016/j.esd.2020.12.002.